

Quantifying heuristic evaluation

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ABSTRACT

Since Nielsen presented his work on heuristic evaluation in the 90s, it has been seen how this inspection method has been used recurrently not only in academia but also in the industry. The benefits of their application have been studied widely and multiple heuristics for multiple purposes have been designed. Despite it, quantifying their qualitative nature has been challenging. This research presents a novel approach to provide a framework for quantifying an heuristic evaluation based on the calculus of what we called Usability Percentage. We have used an initial study that started in 2018 with 7 different evaluators and we presented a new methodology to calculate the aforementioned percentage and also provide multiple information that might be added into a final report. We have used the platform RUXAILAB that allows us to create heuristic evaluations to prepare and execute the test. Our findings are that by comparing both studies the new methodology presents a narrowed data distribution through the different evaluators and allows us to identify in a more precise and concise way which heuristics might fail in an evaluation test.

1. Introduction

In the Human Computer-Interaction (henceforth HCI) discipline, usability evaluation is a key activity to assess the Quality of Use (henceforth QoU) [1] of any interactive system. In this context, Heuristic Evaluation (henceforth HE) [2] has become one of the most successful techniques. A heuristic evaluation can provide valuable insights from multiple perspectives and different scopes [3] with their adherence to established usability principles by HCI experts. This expertise at the same time can act as a dichotomous consequence with both positive and negative implications as it relies the evaluation to a subjective nature relying on the expertise and judgment of the evaluators.

The ambivalent nature of the heuristics have been a challenge for the multiple evaluators that created and answered an heuristic evaluation. To surpass this difficulty, some deterministic quantifying methods that might allow to convert into numerical values the different perspectives of the different evaluators have been created. The purpose of that is leveraging data-driven techniques, to bring objectivity and precision to the analysis of heuristic evaluation [4–10].

The following research introduces a novel quantifying method to evaluate an heuristic evaluation through the refinement of an existing proposal by Granollers [11] where they merged into 15 heuristic principles based on Nielsen [2] and Tognazzini [12]. In total, 60 questions depicted the different heuristic where expert evaluators where in

charge to respond them in 4 different options. The new methodology proposes a novel approach about how to quantify different heuristic answer by a giving set of questions. The main goal was to improve an existing method providing more tools for the experts that allows to properly express their thoughts through the evaluation process. To achieve this we have reproduced the initial study over the same system and also using the same experts to perform the evaluation using the updated methodology. To achieve this, we have developed and used the RUXAILAB system [13] that allows to perform a complete heuristic evaluation exercise. Specifically, it allows to carry out from a classical HE (as defined by J.Nielsen) to the variants included in different proposals found in the literature review done by Quiñones and Rusu [3]. In our case, it allowed us to create a customized heuristic evaluation and configured a novel mechanism to calculate what we have depicted as Usability Percentage. In this study we have expanded the number of possible answers from four to seven and we have updated the method to calculate the Usability Percentage to compute an average method and also upper and lower deviations depending of the questions answers.

The remainder of this study is organized as follows. Section 2 presents the background related to heuristic evaluation and different methods that have been used to quantify an heuristic test. In Section 3,

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we explore the base work that we have been used to expand our research. We present the methodology used to calculate what is called by “Usability Percentage” and describes the experimentation that was made discussing about the results and highlighting the concepts that will be taken into account for the new methodology. Our novel methodology is depicted in Section 4, where we describe it from two different perspectives; the first one from the point of view of the changes on the questions that answers an heuristic and the second one to the formula used to calculate the Usability Percentage. Moreover, it is presented an experimentation procedure used to properly compare the previous methodology and the new one. Finally, the conclusions and future work are presented in Section 6.

2. Background

Usability is defined by ISO 9241-11 as the “extent to which a product can be used by specific users to achieve objectives with effectiveness, efficiency and satisfaction in a specific context of use” [14]. There are multiple methods to evaluate or measure usability, one of them is called Heuristic Evaluation, which is an inspection method that helps to identify problems based on a set of heuristics and principles [2]. In the below section, we present the concept of HE and discuss the quantification of this method, studying the literature and the different approaches and methodologies presented by the authors with related work.

2.1. Heuristic evaluation

Heuristic Evaluation is an usability inspection method that was first introduced by J. Nielsen and R. Molich in 1990 on their work titled “Heuristic Evaluation of User Interfaces” [2]. In this article their describe a usability evaluation method where expert evaluators assess the quality of use of an interactive system based on inspecting its User Interface (UI) by checking the degree of compliance of a set of a predefined usability guidelines or heuristics (hence the name). This technique is based on a chosen list of guidelines and, by the experts knowledge, where their experience may help to understand and identify in an effective way usability problems or enhancements. Their method provides a set of 10 heuristics to be used during the evaluation, first the experts inspect independently the interface and prepare an individual list of the problems encountered, then they gather with other evaluators to have a discussion and aggregate their answers, to find out a more cohesive final list. By the end of the discussions, evaluators rank the problems taking into account the scales of severity, frequency and critical aspects of the items found.

Performing heuristics evaluations have various advantages (1) it is cost-effective as it requires neither much time nor a lot of users compared to traditional user testing; (2) it does not require a demanding planning, less people involved and smaller procedure of analysis; (3) it is applicable in multiple phases of software, being able to be applied on planning, development and on ready-to-use products as well [2,15]. Despite the technique might be considered outdated for the time when it was created, the aforementioned topics suits perfectly well with the industry standards and software development processes. Any kind of agile methodology benefits from apply it heuristic evaluation at the initial stages of the development to improve the quality of their products as if allows evaluators to quickly pinpoint problems in environments where time and resources are limited. Moreover, their versatility allows the method to tailor almost every purposes as it is able to adapt across different domains.

Nevertheless, there are also disadvantages, such as: (1) it is hard to encounter evaluators that are experts, and will be able to give good quality feedback; (2) depending on project stage, evaluators might have a hard time to understand what are the possible tasks they can perform on the software; (3) the results may not have suggestions on how to resolve the usability problems found, which could require an extra data

collecting [2,16] and, finally, (4) original punctuation from Nielsen and Molich [2] has traditionally been confusing for the evaluators by scoring the three different attributes [17] (severity, frequency and criticality), causing that most of the experts end up using only one attribute [3,18–20] (usually criticality) being and important argument why many of the subsequent proposals have focused on improving this part.

The initial set of heuristics created by Nielsen and Molich were considered too general [21], opening up new studies to improve the initial set to be more specific to the desired object of study and also the proposal of new sets, such as the well known Shneiderman’s eight golden rules [22], Norman’s seven principles [23] or Tognazzini’s first principles of interaction design [12].

Many authors have proposed new groups of heuristics specific to their use cases which led to the investigation and development of methodologies to create this data. The work done by Quiñones et al. [3] presents a systematic review of different usability heuristics proposals that have been done in literature. Their results presents different approaches to develop new heuristic sets taking into account existing heuristics, methodologies, literature reviews, based on usability problems, mixing processes, design recommendations, interviews or theories. After analyzing more than 70 papers they concluded that there are two clusters for heuristic research, one based on studies related to developing usability heuristics for a specific domain and another to studies related to processes or methodologies to develop usability heuristics.

Regarding the process of creating new heuristics, other authors have studied the existence of a consensus about the most effective process [4–8,10,24], where most studies had in common the exploration of literature, comparative analysis of methodologies and the extensive experience of researchers [5].

2.2. Quantification of usability

Heuristic evaluation has started as a qualitative evaluation to find out problems on a desired interface [2]. Experts would be given a software to analyze and then would give their personal feedback based on their own knowledge and experience. Although, identifying the problem within the feedback of each evaluator without a numeric measurement requires some effort from the researchers to extract metrics such as effectiveness, efficiency and satisfaction [25]. Therefore different techniques to quantify the usability have been developed by authors.

The work of Mitta [26] proposed a methodology for quantifying expert usability by using a linear multivariate function, where user perceptions and performance are the independent variables. The methodology is illustrated through a practical example, where a usability function is derived with variables specified by interface design experts. Linear normalization was used to map not normalized experimental data to a common range, the calculation resulted in the measured usability score.

The quantification of the attributes depicted by Nielsen and Molich [2] as severity has also been studied by Delice and Güngör [17] where aimed to detail usability problems on website pages and use a combination of the Analytic Hierarchy Process (henceforth AHP) [27] and HE to analyze which ones are more critical to the user experience. The method was analyzed by promoting an evaluation over a website, retrieving expert’s feedback and extracting usability problems. Those problems would then be ranked according to a pairwise comparison following the AHP methodology, weighting following the Saaty’s 1-9 scale [27].

In the other hand, Granollers [11] researched the combination of different methods to perform usability evaluation, combining the proposal of a 15 principle heuristic set with specific questions to be answered. To quantify the answers, a 4-option rating scale was created,

each answer having a numeric value attached to the label. The quantified representation was called as Usability Percentage (henceforth UP), first it was calculated the percentage of each evaluator and then the final UP was retrieved with the means of all of the participants. This methodology was collected as a notable proposal in the book “Interaction Design: Beyond Human–Computer Interaction by the authors Preece, Jenny, Rogers, Yvonne, Shard and Helen”. [28].

Paz et al. [29] proposed the adoption of specialist-orientated inspection technique to quantify usability, giving users a 64-item checklist, elaborated as Yes/No questions grouped into six categories, inspired by the group of HE to analyze UX on websites validated by Bonastre and Granollers [30]. In their work the usability was quantified by calculating the average of the evaluators answers in each website, and they observed that the scoring was uniform, therefore the assessment methodology was a reliable tool to evaluate usability. Regarding the checklist we can also find the work done by [31] where they provide an extension based on Nielsen’s list where it is possible to answer with “Yes”, “No” and “N/A” for those answers that do not apply. However those results stands as individual and do not provide a general insight for the heuristic or the test itself. Regarding studies that analyze the mean of the results and other statistical variables we can find the work done by Khowaja et. a. [32] where they prepared a checklist of 25 heuristics and graded their expert responses into a numerical range from 0 to 4 where 0 corresponds to “No violation” response and 4 to “Severe” responses. As they pointed it helped to provide data from each heuristic and create a severity rating that helped them to analyze the medical tool that was used on their research.

The use of a checklist is something common on the research [11, 29,32–34], however some studies do not provide details or experimentation where the checklist was used [9,33,35]. Another topic of interest is the number of questions used during the evaluation. We found that several checklists end up with a higher number of sub-heuristics or topics to be answered by the evaluators. For example, the work done by Yanez et al. [33] uses 230 sub-heuristics, 436 are used on the work done by Khowaja et al. [32] or 460 on the work done by Omar, et al. [35] are some examples of heuristic evaluations with a large number of questions to be answered by the evaluators. As Nielsen pointed, a high number of answers could end in a perceived exhaustion on the part of the evaluator [2]. We do not find data regarding this point of view on the studied papers but on the work done by Granollers [11] it was established a maximum of 60 sub-heuristics, formulated in terms of questions to be answered by the evaluators, after merging two heuristics sources and they note that they were trying to reduce the number of questions to avoid the aforementioned problem. Another problem that arises after analyzing different heuristic studies is that each one is done in different supports. Some papers provide a PDF file [31] where the checklist items to be printed while others provide documents with “doc” [32] extension or others with “xls” extension [11].

2.3. RUXAILAB

Traditional evaluation methods are still widely used by many researchers [7], like the Nielsen’s heuristics for example, one of the most used set of heuristics [3]. These tests are mostly performed on usability labs with specialized instructors and equipment [36]. But with the rise of the new technologies, interaction methods and interfaces have evolved, along with that, the techniques to evaluate those interfaces would also have to follow the tendency.

In the context of usability evaluation, multiple software systems have appeared to help performing usability evaluations. Those tools allows to perform activities like user testing, card sorting, thinking aloud or heuristic evaluation [37]. However as it was studied before by Capdevila et al. [37], most tools are expensive and have limited free tiers. In order to offer an accessible, free and open source solution,

the RUXAILAB project¹ was created to serve as a remote usability laboratory [13].

The software consists of a web application designed as a facilitating tool for applying usability evaluation through inquiry, inspection, and testing methods [13]. The application allows researchers to share their studies with others using a standardized and simplified framework. This feature enhances communication between researchers and evaluators and aids in disseminating knowledge within the academic field in an accessible manner. Currently, the platform supports the creation of usability experiments by allowing users to choose their type (as of now, only User Test and Heuristic Evaluation are implemented) and invite collaborators, such as other researchers and evaluators. System users can modify the values of their heuristics as needed, enabling them to update these values alongside their experimentation and formula validations. The tool also assists with data analysis by generating study reports containing multiple metrics, which can be exported if needed. The open-source nature of the application has a significant impact, as users can modify the framework as desired, updating formulas, analysis methods, and visualizations.

For the heuristic evaluation, the design functionality was based on literature, primarily using the principles defined by Nielsen [2] and Granollers [11]. This consists of a section where users can define their custom heuristic sets. The process of creating heuristics on RUXAILAB offers several options, one of which is the ability to define sub-criteria (or questions) for each heuristic. This method elucidates the criteria-answering methodology, as described by the 44-checklist by Mi et al. [38] and the study by Granollers mentioned earlier. Each heuristic created can have any desired number of questions, and users can customize it to allow multiple-choice options with any desired values. Fig. 1 illustrates how RUXAILAB manages the pipeline of operations for a heuristic evaluation. Sub Fig. 1(a) exemplifies the heuristic creation screen, which contains several options for adapting the heuristic evaluation to the specific purpose of the evaluation. Sub Fig. 1(b) shows the test answer form, which is a straightforward process that the expert evaluator can complete synchronously or asynchronously, with the capacity to add comments or screenshots. Finally, sub Fig. 1(c) showcases how the system gathers and presents the data, offering multiple tables and charts to visualize the experimental results.

3. Previous work

The work presented in this section is based on the study conducted by Granollers [11]. It describes the methodology used in their experiments and discusses how this proposal can be improved.

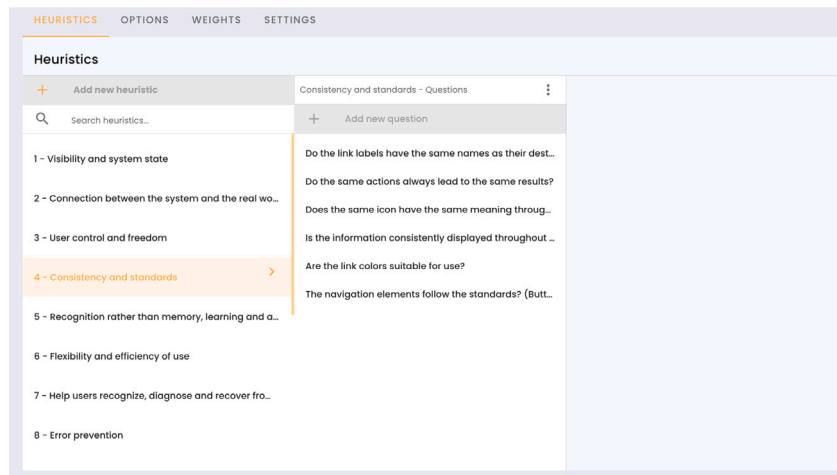
3.1. Methodology and experimentation

To validate the aforementioned proposal it was designed an experiment with 7 evaluators with valuable experience doing heuristic evaluation. The study was done on 2018 and it was applied to the website of the University of Lleida faculty.² Table A.1 on Appendix shows the 15 principles used with their respective 60 questions distributed heterogeneously distributed among the heuristics.

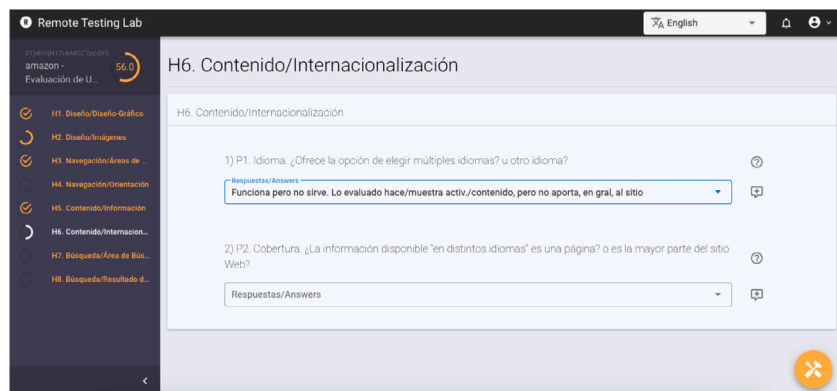
The different answers and values can be seen on Table 1 and after answering them, the rating scale computes what they called the Usability Percentage as a final value. For the answers with values of “Yes”, “Neither yes, nor no” or “No” a numerical value has been added which is later used in the compute of UP. For those values with the result of “Not applicable” they are not considered in the equation, for this reason we split our answers between countable and non-countable.

¹ RUXAILAB’s GitHub repository. Available at: <https://github.com/uramakilab/remote-usability-lab>. Accessed date: August 5, 2024.

² Escola Politècnica Superior. Website: [EscolaPolitècnicaSuperior\ndot\\$UniversitatdeLleida](https://www.espoli.udl.cat/). Last access: August 5, 2024.



(a) Heuristic creation page



(b) Heuristic answering page



(c) Analysis and evaluation page

Fig. 1. Screenshots taken from RUXAILAB web application system for Heuristic Evaluation.

Table 1

Answers options for the given heuristics.

Option	Value	Description
Yes	1.0	Appropriate answer when what is being evaluated is coherent
Neither yes, nor no	0.5	Answer when something is true, but not completely
No	0.0	Appropriate answer when what is being evaluated is not coherent
Not applicable	–	When the question made do not belong to the object analysis scope

Table 2
Distributions of the answers by evaluators on the previous work.

Answer	EV1	EV2	EV3	EV4	EV5	EV6	EV7	AVG	STD
Yes (Y)	44	34	33	39	38	37	29	37	4.82
Neither yes, nor no (YN)	0	1	3	2	3	0	2	2	1.27
No (N)	13	15	3	4	12	6	12	12	5.06
Not applicable (NA)	3	10	21	15	7	17	15	15	6.21
Countable questions (Y& YN& N)	57	50	39	45	53	43	45	45	6.21
Non countable questions (NA)	3	10	21	15	7	17	15	15	6.21

Table 3
Summary results of study by heuristics and evaluators.

Heuristic	Eval. 1	Eval. 2	Eval. 3	Eval. 4	Eval. 5	Eval. 6	Eval. 7	AVG	STD
H1	5	5	4	4.5	4	5	4	4.50	0.50
H2	3	2	4	4	3.5	4	2	3.21	0.91
H3	3	2	3	2	2	1	1.5	2.07	0.73
H4	5	4	5	5.5	4	4	3.5	4.36	0.80
H5	5	4	4	5	5	5	4	4.57	0.53
H6	5	3	3	6	3	5	3	4.00	1.29
H7	3	3	0	0	2	2	4	2.00	1.53
H8	2	2	2	0	2	2	1	1.57	0.79
H9	4	3	4	3	4	4	1	3.29	1.11
H10	0	0	0	0	0.5	0	0	0.07	0.19
H11	2	0	0	1	1	0	0	0.57	0.79
H12	4	3	2.5	4	4	2	2	3.07	0.93
H13	2	2.5	2	3	3	3	2	2.50	0.50
H14	0	0	0	2	1	0	2	0.71	0.95
H15	1	1	0	0	1	0	0	0.43	0.53
N.A.	3	10	21	15	7	17	15	15	6.21
SUM.	44	34.5	33.5	40	39.5	37	30	37	4.69
UP	77.2%	69.0%	85.9%	88.9%	74.5%	86.0%	66.7%	78.32	0.09

UP can be represented by formula (1) which represents the sum of all the countable answers to give a quantitative insight like other studies [11] in terms of quantifying how usable the evaluated system is.

$$UP = \frac{\sum_{i=1}^M (Score_i) + 0.5 \cdot \sum_{j=1}^N (Score_j)}{M + N + X} \quad (1)$$

Where:

UP represents Usability Percentage as the mean score of the responses

Score: corresponds to the score of response i, j . A “Yes” response is assigned a score of **1.0**, a “No” response has a score of **0.0**, and a “Neither yes nor no” response contributes **0.5** to the **Score_j**.

M is the total number of “Yes” responses.

N is the total number of “Neither yes, nor no” responses.

X is the total number of “No” responses.

Table 2 displays the total answers by type for each evaluator, and also the sum of the non-applicable questions and countable questions. While Table 3 represents the findings after evaluating the website with 7 different evaluators. In that Table it can be found the different scores that each evaluator gave for the respective heuristics explained in Table A.1.

From the perspective of the heuristics results it is also represented in Table 3 information about the average value obtained in the evaluation in numerical value and also in percentage and finally the standard deviation.

3.2. Discussion

On two approaches. The first is the way to score the answers, trying to provide a better way to reflect the proper reality, and second, to improve the way how the UP value is defined.

The previous experimentation showed that the maximum UP percentage was 88.9% obtained by evaluator 4 and the lower 66.7% obtained with evaluator 7. We also identified with an average value of 78.31%. Regarding the total number of answered questions with countable values we detected that from the original 60 answers the evaluators answered an average of 47.71 questions, being the evaluator 1 the one who answered the greatest number of answers (57 questions answered and 3 as “Non applicable”) and the evaluator 3 who has answered the least (39 questions answered and 21 as “Non applicable”). Table 3 also show the relation with countable and “Not applicable” questions. In that case it is observed that H10 is the heuristic with the higher number of “Non applicable” answers while H1 the opposite.

Different aspects arises from the previous experimentation regarding the UP not only from the quantitative perspective but also from the qualitative. It was performed a focus group among the different evaluators to extract information about the whole process: the methodology, the use of the new set of heuristics, the questions, the answers and the UP calculation.

The focus group methodology facilitated an in-depth exploration through open-ended discussion, crucial for capturing the intricacies of participant’s opinions and experiences. This dynamic procedure enabled the experienced participants to foster a rich discussion through two different approaches, the set of heuristics, the evaluation method and finally the way how it is computed the usability percentage.

The initial focal point involves an exploration of participant’s opinions concerning the introduced set of 15 heuristics. This analytical approach is substantiated by existing literature, where Nielsen’s heuristics are recognized to be one of the most well-known and used heuristic

list [2], and Tognazzini's list is acknowledge for its precision and comprehensive nature [12].

There were an unanimous consensus among evaluators, pointed from focus group interactions, that the new set of heuristics signifies a noteworthy advancement [11]. This statement comes from affirming that, although the new list presented has a slightly augmented scope compared to Nielsen's list, it is more complete in terms of completeness bringing the Tognazzini spectrum of heuristics. Such assertions within the focus group discussions highlighted the qualitative validation of the heuristics' efficacy, aligning with established usability evaluation literature [39].

The first critical dimension under analysis pertains to the examination of the questions associated with each heuristic, constituting an indispensable element in the evaluation of usability principles. The assertion is underscored by the acknowledgment that the assessment of any individual heuristic is inherently intertwined with the questions devised to operationalize it. This perspective aligns with foundational literature emphasizing the interdependence of evaluative criteria and usability principles [40]. In that case, it was presented only 4 different options to evaluators, simplifying the quantitative process translation to: "Yes", "Neither yes, nor no", "No" and "Not applicable".

For the evaluators perspective this answers had both benefits and disadvantages. On the one hand, the fact that the range of answers were simplified helped to reduce the bias and provides more unanimous consensus through evaluator and through heuristic. It can be seen on Table 3 how scores on heuristic 10, 11, 14 or 15, evaluators provides similar answers having lower values. Moreover, it helps on the burnout [2] of having to answer multiple questions with different dimensionality as heuristics are. This simplification causes, on the other hand, the disadvantages of the proposal. When qualitative data were analyzed during the review process on the focus group, it was noticed that the comments for those heuristics that has lower values indicates that in most of the cases they do not know how to correctly proceed in terms of giving an appropriate answer.

Moreover, we have also encountered difficulties on the understanding or discern between "Yes" and "Neither yes, nor no". In absolute terms, "Yes" answer was designed to give a clear answer on those responses that were irrefutable and always present through the evaluation. However "Neither yes, nor no" adds in their description that must be the accurate answer when although being true in most of the cases if it is not completely true must choose this answer. It implies that the evaluator has to remember through the complete evaluation of the corresponding question that if it detects something that is not present in the evaluation, the answer "Neither yes, nor no" option must be considered. Regarding the answers of the evaluators, it was sometimes difficult to remember because in some cases they had to review some answers again because they identified new flows that they had not considered previously.

4. Enhanced methodology

A novel methodology is presented in the following section based on the previous study realized and taking into account the valuable information extracted from different sources that have used this proposal [29,34].

In that sense we offer different alternatives based on two approaches. The first one will be based on modifying the quantification process of the answers that can be given on the heuristic questions and the other will be based on how the UP is calculated considering the new dimensionality of possible answers.

4.1. New set of answers

As it was depicted by the evaluators, one of the benefits of the proposal was the simplifying process of answering a set of questions to determine an heuristic value. However, this restricted set does not allow to correctly map the diversity of answers. This ended in various factors that allows to explore the increase of the bias for the options that may be offered to the evaluator to answer the heuristics.

Moreover, it was identified that the initial set of possible answers has a non countable value called "Not applicable". Different evaluators indicate that this term was ambiguous and could lead to an error of interpretation. Also, it was identified that some questions were impossible to answer as they could not find an answer for it. For example, question 3 from Heuristic 10 regarding **Help and documentation**, asks for "Is the help section aimed at solving problems?". In that case, most of the evaluators do not know how to produce an error on the system so they could not evaluate if an existing help section may help or not to solve an unencountered problem.

To give more flexibility, we have separated the answers in two kinds. Those that have a value and will be countable in terms of modifying the UP, and the other values that do not change but they are also valuable in terms to know how the heuristic is being evaluated.

Table 4 shows the different answers divided in two blocks. On the one hand, we have those results that has a value number which are the updated version of "Yes" or "No" answers. And from the other hand those values without a value. Every option is provided with examples for further understanding. Important to point the last answer which is called "Impossible to check", this answer will be useful on those cases that the evaluator do not know how to answer because there is no trace on the evaluated system to know or understand how to accomplish or answer the question.

4.2. New compute of UP

In Section 3.1 it was presented the formula to compute the UP based on four different answers. However, the introduction of new variables slightly changes the original formula which is depicted on the Formula (2).

$$UP_{\alpha} = \frac{\sum_{i=1}^M (Score_i) + 0.6 \cdot \sum_{j=1}^N (Score_j) + 0.3 \cdot \sum_{k=1}^O (Score_k)}{M + N + O + X} \quad (2)$$

Where:

UP_{α} represents Usability Percentage as the mean score of the responses.

M is the total number of "Yes, in All cases" responses.

N is the total number of "Yes, but some cases missing" responses.

O is the total number of "Not always" responses.

X is the total number of "No, in NO case" responses.

Score: corresponds to the score of response i, j, k . A "Yes, in All cases" response is assigned a score of 1.0, a "Yes, but some cases missing" response has a score of 0.66, a "Not always" response has a value of 0.33. A "No, in NO case" contributes 0.0 to the score, so it is not represented in the formula.

The computation of UP_{α} only includes the numeric values represented by Table 4. Other values are not considered in the formula. However, in the case of the questions answered by the "Impossible to check" option, we will adapt the formula to display the potential improvement or losses in terms of average. We consider the introduction of a variable Z depicted in Eq. (3). It allows a seamless transition between two scenarios. When $Z = 1$, the formula corresponds to UP_{β} , incorporating the influence of conditional instances ($Score_l$).

Table 4
Different answers and their corresponding values for the new methodology.

Option	Value	Description
YES, in ALL case	1.0	Appropriate answer when what is being evaluated has been fulfilled in all checked cases
Yes, but some cases missing	0.66	Appropriate answer when what is being evaluated has not always been fulfilled, but in most checked cases.
Not always	0.33	Appropriate answer when what is being evaluated has practically not been fulfilled in any checked case.
NO, in NO case	0.0	Appropriate answer when what is being evaluated has not been fulfilled in any of the checked cases.
Not applicable	–	Appropriate answer when what is being evaluated is not found on the website, but is not relevant.
It is NOT a problem	–	Appropriate answer when what is being evaluated is not a problem.
Impossible to check	–	Appropriate answer when what is being evaluated could not be fulfilled

Conversely, when $Z = 0$, the formula simplifies to UP_γ , excluding the conditional term and providing a baseline usability percentage calculation without considering the sum of the responses of $(Score_i)$.

$$UP_i = \begin{cases} UP_\beta & \text{if } Z = 1 \\ UP_\gamma & \text{if } Z = 0 \end{cases} \quad (3)$$

After describing this new approach we update the formula (2) to express the “**Impossible to check**” values and their both resulting options denoted by UP_β and UP_γ .

The unified formula is expressed as:

$$UP_i = \frac{\sum_{i=1}^M (Score_i) + 0.6 \cdot \sum_{j=1}^N (Score_j) + 0.3 \cdot \sum_{k=1}^O (Score_k) + Z \cdot \sum_{l=1}^P (Score_l)}{(M + N + O + X) + P} \quad (4)$$

Where:

P is the total number of “**Impossible to check**” responses.

After describing how to calculate the new percentages, we introduce a new equation to represent the percentage difference between UP_α and their variants of profit UP_β and losses UP_γ . We define the percentage difference (Percentage Difference _{i}) as the relative change between a variant scenario (UP_i) and the original scenario (UP_α). The formula is given by:

$$\text{Percentage Difference}_i = \left(\frac{UP_i - UP_\alpha}{UP_\alpha} \right) \times 100 \quad (5)$$

This formula expresses the percentage difference as a proportion of the original value, providing insights into the change in performance between the original scenario and the variant scenario represented by i . The subscript i can take values such as β or γ depending on the specific scenario being considered.

4.3. Experimentation

To compare correctly the previous methodology presented by Grannollers [11] and explained in Section 3, we have replied the same study with the same evaluators but considering the new methodology presented on Section 4. Participants were submitted to the same questionnaire, and could choose between the 7 new different answers, which were presented on Table 4. It was also used the same website that was used since it has not been altered since it was done the initial evaluation. We have followed the same steps depicted on the original paper [11] but instead of using an excel sheet with XLS extension we have used the open source software aforementioned on Section 2.3 to centralize the test information and also evaluators answers. As it was described the system is capable to count the different answers since we have complete freedom to create our own evaluation system by attributing weights to the heuristics and their underlying questions.

Moreover, we can use the system to send invitations to our collaborators through the platform and save the session to be completed in different periods of time.

To apply the new formulas, we needed to gather required information first, so after their answers, it was summed the occurrences of each type of answers by evaluator. It was also gathered the total of countable and non countable questions, which are considered the ones with label of “**Not applicable**” and “**It is NOT a problem**”. These values are needed for variables $Score_{i,j,k,l}$ and M, N, O, X, P .

To be able to compare the methodologies, we calculated UP_α , UP_β and UP_γ by using the formulas (2) and (4) presented in the previous section. It was also calculated the Percentage Difference (PD) to find out how much it varies between the different methods.

Table 6 presents a summary of the study results. The first section of the table shows the sum of answers by evaluator on each heuristic, which was calculated taking into account the countable answers presented on Table 5. It can also be seen the average results and the usability percentage by heuristic, considering the standard deviation. The table also presents the total of “**Impossible to check**” answers, represented by the ITC. and the “**Not applicable/It is NOT a problem**” answers represented as NA. The collected data resumed on the first part of the table are required to calculate the UP and PD variants using the formulas (2), (4) and (5) mentioned before.

5. Results

In the following section, we analyze the obtained data and discuss it regarding the updated methodology also comparing the results from the original method trying to provide a general overview of their benefits. We have split this section into different subtopics regarding different perspectives analysis. We start analyzing the new set of possible answers and how they have influenced the compute of the results. We also analyze the values obtained from the different heuristics and study their implications after applying the updated methodology. Moreover, we analyze the difference between the calculation of the UP values and depict how UP_α , UP_β and UP_γ might be useful for the evaluation responsible. Finally, we present some insights received by our evaluators after applying a focus group session at the end of the heuristic evaluation.

5.1. From the perspective of the answers to the questions

In Table 5 we collected the information regarding the different evaluators and their scores from their respective answers. For the response “**Yes, in All cases**”, the EV1 has been the one who has obtained the higher number of responses, 29; while EV7 is the one who has obtained the least, 20. Regarding “**Yes, but some cases missing**”, the evaluator with the higher number it has been EV4 with a total number of 10, while the lower has been the EV5 with 5. The final countable answers “**Not always**” has obtained 8 answers from EV2 and

Table 5
Distributions of the answers by evaluators.

Answer	EV1	EV2	EV3	EV4	EV5	EV6	EV7	AVG	STD
Yes, in ALL cases (Y)	29	27	24	27	28	25	20	25.71	3.04
Yes, but some cases missing (Y-)	9	6	6	10	8	5	6	7.14	1.86
Not always (N+)	5	8	3	6	4	4	4	4.86	1.68
No, in NO case (N)	6	8	5	2	8	5	6	5.71	2.06
Not applicable (NA)	1	9	15	8	6	13	7	8.43	4.61
It is NOT a problem (NP)	5	1	4	3	2	4	4	3.29	1.38
Impossible to check (ITC)	5	1	3	4	4	4	13	4.86	3.80
Countable questions (Y & Y- & N+ & N)	49	49	38	45	48	39	36	43.43	5.62
Non countable questions (NA & NP & ITC)	11	11	22	15	12	21	24	16.57	5.62

Table 6
Summary of experimentation results by evaluators and heuristics.

Heuristic	EV1	EV2	EV3	EV4	EV5	EV6	EV7	AVG	STD	HUP
H1	4.32	4.66	3.66	4.32	3.66	3.99	4.33	4.13	0.38	82.69%
H2	2.66	2.66	3	2.33	2.99	2.64	1.32	2.52	0.58	62.93%
H3	1	2.66	3	1.65	2	0.66	1.33	1.76	0.86	58.57%
H4	5	4.99	4.32	5.32	2.99	3.66	3.66	4.28	0.87	71.29%
H5	3.64	4	3.66	3.99	3.98	4.66	3.66	3.94	0.36	78.83%
H6	3.99	3	3	5.66	3.33	5	2	3.71	1.27	61.86%
H7	3	2.3	0	0.33	1	0	3	1.38	1.36	34.39%
H8	2	1.66	1.32	0	2	2	1	1.43	0.74	47.52%
H9	4	3.3	2.66	2.99	4	3	1.33	3.04	0.91	76.11%
H10	0	0	0	0	0.3	0	0	0.05	0.12	0.94%
H11	1.66	0	0	1	1	0	0	0.52	0.69	17.43%
H12	4	1.32	2.33	3	3.32	1.66	1.99	2.52	0.96	62.89%
H13	0.99	1.99	2	3	3	2.3	1.66	2.14	0.72	71.29%
H14	0	0	0	2	1	0	0	0.43	0.79	14.29%
H15	0.33	1	0	0	0	0	0	0.19	0.38	9.5%
I.T.C.	4	1	1	3	3	4	13	~4	4.1	–
N.A.	6	17	20	10	14	18	18	~12	4.68	–
SUM.	36.59	33.57	28.95	35.58	34.6	29.62	26.30	~32	4.14	–
UP α	76.23%	69.94%	76.18%	79.07%	72.08%	75.95%	70.22%	74.24%	3.49	–
UP β	78.06%	70.55%	76.79%	80.38%	73.73%	78.19%	78.12%	76.54%	3.32	–
UP γ	70.37%	68.51%	74.23%	74.13%	67.84%	68.88%	51.59%	67.94%	7.66	–
PD β	2.34%	0.87%	0.79%	1.63%	2.23%	2.86%	4.97%	2.97%	3.24	–
PD γ	-7.69%	-2.04%	-2.56%	-6.25%	-5.88%	-9.30%	-26.53%	-8.60%	8.32	–

3 from EV3 as their higher and lower values respectively and regarding the answer “**No, in any case**”, the maximum value was achieved by the evaluator EV2 who considered 8 questions with that answer, and the lowest values came from the EV4 with 2 answers. The last 3 possible answers, “**Not applicable**”, “**It is not a problem**” and “**Impossible to check**” have received from EV3, EV1 and EV7 their maximum values, 16, 5 and 13 respectively. While their lower values came from EV1 on “**Not applicable answer**” and EV2 for “**It is not a problem**” and also “**Impossible to check**”. In total the evaluator that has chosen the higher number of Countable questions is the EV1 and EV2, while the EV7 the opposite. In that line the EV7 has been who has chosen the higher number of non-countable questions while the EV1 and EV2 the least. Comparing this results from those obtained on Table 2 we can see similitudes between some results across evaluators. For example, EV1 and EV7 has maintained their position as a higher and lower rank for the highest computable question which was “**Yes**” on the initial study and “**Yes, in All cases**” on the new one. The same happened with the “**Not applicable**” answer which followed the same pattern for EV3 who obtained the highest value and EV1 the lower. This outcome may stem from the recent redistribution of results, as both answers exhibit wide quartile ranges. Despite adjustments to the values, which have led to a reduction in the final score, the impact has not been significant enough to alter the final result.

On Fig. 2 we have represented the difference between the values obtained from the first study versus the results from the new methodology. As it was described on Section 3, one of the problems encountered

by the evaluators was that, despite the simplicity of the method of having 4 possible answers, sometimes they found it difficult to answer properly the given question. This is the first thing that we can most notice, with an existing difference between the two possible different positive values “**Yes**” and “**Neither yes or no**” with a representation of the 60.48% and 2.62% respectively. With the new system, these values have suffered a slight modification with the inclusion of the new values. We can find an evidence when we observe how the answer “**Yes, in all cases**” have seen their value decreased in favor to the other options with a total value of 42.86%. We can also observe how the answers “**Yes, but some cases missing**” and “**Not always**” have higher values than their predecessor, in that case the values obtained are 11.9% and 8.10% respectively. However, if we compare the sum of the countable values from one methodology to another we can notice a lower difference between the original version with a value of 63.1% and 62.86% from the new one. It means that the addition and modification of the countable answers have helped to modify answers like “**Yes, in All cases**” into the other countable values without modifying the overall ratio obtained in the previous study.

On the other hand, analyzing the negative answers considered as “**No, in any case**” we can also see a notable difference between the original and new methodology. From a value of 15.95% it has been decreased into 9.52%. This fact lies in the introduction of a new set of answers considered as non-countable options that provides also a more flexible way to indicate to the evaluator that a result can be considered as negative in some way. In parallel, this is also related

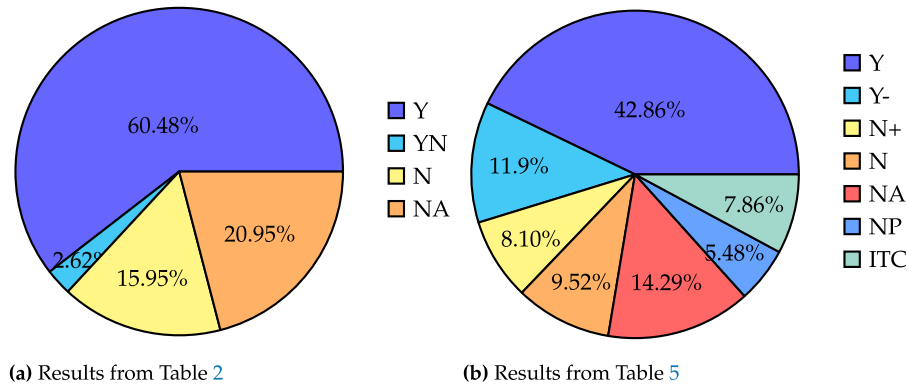


Fig. 2. Pie chart representation of the results available on Tables 2 and 5 after applying the methodology depicted on Sections 3 and 4 by means of the answer type.

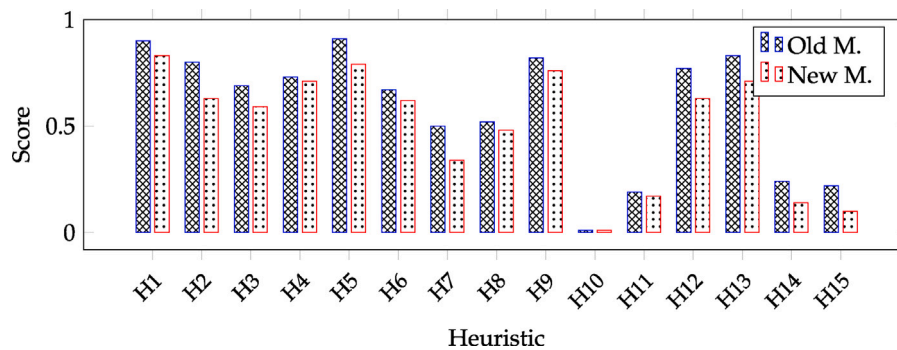


Fig. 3. Comparison of Heuristic Evaluation Results.

to the results obtained from the previous methodology where “Not applicable” answers have a representation of 20.95%. With the new methodology, those values were split into “Not applicable”, “It is not a problem” and “Impossible to check”. In that case, the sum of all of those values has obtained a value of 27.63% with 14.19%, 5.48%, and 7.86% respectively. Which means a difference of almost 7% compared to the previous method. What can be seen on that point is that while in the previous version, most of the answers fell on the “Not applicable” version with the new methodology we have proposed a method to map more adequately the different non-countable answers to understand and provide a better explanation of the heuristic evaluation done by the evaluators.

Finally we have the question of the “Impossible to Check” answers. The inclusion of this last answer might not had a very high representation with a value of 7.86%. Analyzing this result deeply, most of the answers that originally were depicted as “Not applicable” have converted into this new option with a rate of 1/3. However, it seems that this answer also obtained one of the highest standard deviation value ($\sigma = 3.8$) where the EV7 have choose this option for 13 different questions. Compared to the other evaluators, they have not used this option in such a repetitive way. The second evaluator who has used it the most has been EV1 with a total of 5 responses and the average was 4.86, so we could consider that this evaluator has been more critical than his counterparts.

In general terms, the fact that the number of countable and non-countable questions suffered a modifications it might be related to multiple aspects. However it is clear that if we compare the results from Tables 2 and 5 regarding countable and non-countable answers we can see how the number of countable answers have been decreased

in the new methodology as it is illustrated on Fig. 2. We believe that the fact that the evaluators had to evaluate an already evaluated system, it helped them to be more critic with their scores as they already know the interface despite the fact that passed more than 6 months between the original experimentation and the new one. However, the site analyzed, do not suffered modifications on their interface neither their functionalities.

5.2. From the perspective of heuristics

Regarding the heuristics we can find the information on Tables 3 and 6. For better interpretation we present Fig. 3 where we have summarized the different values obtained in the different heuristics from the previous and new methodology. We have normalized data to properly analyze it as it was suggested by Mitta et al. [26] it was a necessary constraint to understand each heuristic by their own. One of the first things that we can notice is that it is clear that after applying the new methodology all the heuristic scores have seen their values decreased. The fact that adding a new dimensionality on the set of countable answers had a direct impact when expert evaluators had to response the questions and it has been seen across the test.

Comparing the initial methodology and the new one, we found differences between the highest value. In the initial methodology the highest heuristic was H5 while in the new was H1. However, the lower value was the same for both methodologies which was H10. If we also analyze the normalized data we calculated the quartiles to understand better the distribution of the results. In that case we found that the first quartile (Q1) has a value of 0.24, the second quartile (Q2) a value of 0.51 and finally the third quartile (Q3) a value of 0.83 from the

previous methodology and a value of 0.17 for Q1, 0.535 for Q2 and 0.71 for Q3 on the new methodology.

These quartiles allow a clear segmentation of the data, indicating the proportion of heuristics that present lower or higher performance in relation to the entire evaluated set. Again we can see that comparing between them Q1 and Q3 have their values decreased from the previous study. To interpret this data we have split the values regarding which heuristics falls into the different quartiles.

1. **Heuristics Below Q1:** 4 of the 15 heuristics evaluated are observed to perform below the Q1 from both previous and updated methodology, suggesting areas of improvement for these specific heuristics.
2. **Heuristics between Q1 and Q2:** Only 1 heuristic fall between Q1 and Q2 from the previous methodology and 2 from the updated version of it, indicating a moderate performance compared to the total set.
3. **Heuristics between Q2 and Q3:** A total of 8 heuristics demonstrate performance above Q2 but still below Q3 from the previous work while on the updated version this value is decreased to 6, highlighting their effectiveness compared to the majority of the heuristics evaluated.
4. **Heuristics above Q3:** Finally, only 2 of heuristics stand out by exceeding the Q3 and 3 from previous and updated methodologies respectively, indicating exceptional performance in relation to the entire evaluated set.

It was also expected that, after applying the new methodology we would see a decrease of the general value of the standard deviation of the different heuristics. This hypothesis where based on the fact that adding a more refined way to answer the positive values will reflect in reduce the spread of the data. Although in the initial methodology we cannot found specifically higher values, we detected that some heuristics have seen their values decreased. For example, H6 and H7 obtained a value higher to 1 point (1.29 and 1.53 scores respectively extracted from Table 3) and after applying the new methodology they still be the heuristic with the data more spread out from the mean (1.27 and 1.36 scores respectively extracted from Table 6). For most of the heuristics the new standard deviations values are lower (H1, H2, H5, H6, H7, H8, H9, H10, H11, H14 and H15) representing the 73% of the heuristics indicating the expected less variability or data dispersion of the values compared with the initial methodology. However for the heuristic H3, H4, H12 and H13 the values are higher and the dispersion of the results in that case have been increased although that for H4 and H12 this difference has been less than 0.1 points which implies that it might be possible that if only one question was answered differently those heuristic could have fallen into the initial group raising the value to 83% of heuristics with a reduction of spreading values.

5.3. From the perspective of the usability percentage

Another novel contribution of this research it has been the addition of a new way to compute the UP for the heuristic evaluation. Although it was said on [11], “this percentage must not be taken strictly as a full usability meaning, but as an orientation”. However, some authors have been looking to quantify how usable is a system [3,30,41–43]. We also have expanded this compute to every heuristic which we denoted as Heuristic Usability Percentage (henceforth HUP) to understand how representative is a heuristic and how it can contribute to the final compute of the UP global percentage. Fig. 4 is a visual representation of the data obtained in Table 6 that uses box plots to represent the different values of UP_α , UP_β and UP_γ . We have also added the original value from the previous study denoted by UP.

One of the first things that we can notice is that the UP box has a higher dispersion compared to the new values proposed in this study. This is the widest Interquartile Range (heinfence IQR), spanning

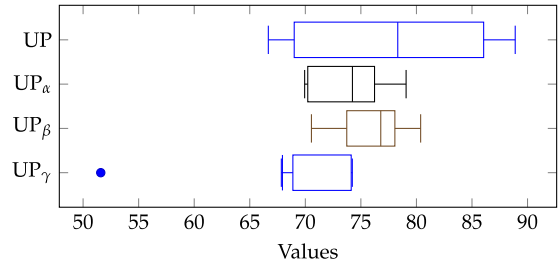


Fig. 4. Boxplots representing the usability percentages (transposed).

Table 7

Final usability percentages with PD_β and PD_γ values.

Option	Value
EV1	76.23% $\pm$ (2.34% -7.69%)
EV2	69.94% $\pm$ (0.87% -2.04%)
EV3	76.18% $\pm$ (0.79% -2.56%)
EV4	79.07% $\pm$ (1.63% -6.25%)
EV5	72.08% $\pm$ (2.23% -5.88%)
EV6	75.95% $\pm$ (2.86% -9.30%)
EV7	70.22% $\pm$ (4.97% -26.53%)

17.05%. This broader range suggests that in the previous methodology, the set of answers was more diverse which is something that we already saw in Section 5.2. However, if we compare it to the other scenarios where UP_α , UP_β and UP_γ are presented it can be seen that they present a narrower IQR range (6.01%, 4.77% and 5.25%, respectively). We can conclude that this is an indicator of a more consistent range of the answer dispersions which can end in a better optimization of the results given by our evaluators.

We can also observe the existence of an outlier within the dataset on the set of values of UP_γ . We have calculate it using the following formula which is the standard method to calculate lower bound for outliers:

$$\text{Lower Bound} = Q1 - 1.5 \times \text{IQR} \quad (6)$$

Indeed this Lower and Upper Bounds were calculated for all the datasets, however only on UP_γ was encountered an outlier. The choice of 1.5 comes from the common convention from the context of Tukey's fences and boxplots [44]. On that case and if we compare it with the results from Table 6 we can see how EV7 was very critic compared to the others evaluators. It has been the evaluator with the higher number o ITC answers with a value of 13 which ends with the higher PD_β and the lower PD_γ values compared with the rest of the evaluators.

Finally, and regarding PD_β and PD_γ , we suggest to use it instead of the UP_β and UP_γ values in order to share the results and compare it among the evaluators. We have shared both information to our evaluators and we only have qualitative data that verifies this suggestion which is analyzed on Section 5.4. In general terms, the authors expected thought this study to provide a more enriched response for the UP, so adding the percentage difference inside the concept of UP has been seen as something valuable that must be considered. Table 7 provides the final list with the suggested UP for the different evaluators after the implementation of the methodology presented on Section 4.

5.4. From the perspective of evaluators

Finally, the last dimension to explore is the qualitative contribution of our different evaluators. As it was explained, seven evaluators performed our board of experts and they were the same for the first and the second experiment. After all of the evaluators answered all of the



Fig. 5. Heatmap representation of heuristic answers by evaluator. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

questions we scheduled a focus group to analyze the overall evaluation and obtain more data. In that sense, the platform RUXAILAB helped as a gathering point of information allowing not only to save all the responses and obtain the quantitative data but also to store comments in the form of text and images from the different evaluators when they answered the evaluation. This platform, also provides a visual representative heatmap of the responses, crossing the information by heuristic and by the evaluator. Fig. 5 shows how this data is represented and uses a color ladder from red to green for the lower and higher values respectively. This figure was used by the evaluators and they find it very useful and helped them to get the data without having to enter into the details and understand the numbers. If we compare both figures when each one represents a methodology we can find that there is no much visual difference and what can be extracted is the same information that we have analyzed on the previous sections that is that heuristic 10, 14 and 15 obtained the worst results and EV7 has been the most critic evaluator. However, as it was depicted, all of the evaluators found it very useful in terms to have a general overview and a fast lecture of data. We have to notice that this tool might be more useful to compare results from different expert evaluators, or different systems analyzed, helping to understand if some heuristics are failing or if exists some kind of bias between expert evaluators.

Regarding the focus group and during the session, we forced the evaluators to discuss the heuristic methodology itself, trying to obtain positive and negative comments about it. Table B.2 on Appendix B shows experts' quotations that are related to the new methodology. We codified every quote into smaller topics for better comparison among them. We have summarized them for better understanding and counted how many times appeared that message and also their categorization if the comment has a positive or negative meaning (see Table 8).

In general we have encountered both positive and negative comments from 4 different topics: "Not complicated/Hard to understand", "Used the new options/Not used the new options", "Useful/Not useful", "Faster/Slower". Topics "Precise" and "Not necessary" we have not considered that one is a counterpart to the other. We can also see a clear tendency to positive comments versus the negatives ones, in that case we have counted 23 positive comments and 9 negative.

Table 8

Similar comments made by evaluators.

Comment	Times mentioned	Statement
Not complicated	2	Positive
Used the new options	6	Positive
Useful	6	Positive
Precise	5	Positive
Not necessary	1	Negative
Not used the new options	1	Negative
Faster	4	Positive
Not useful	1	Negative
Hard to understand	3	Negative
Slower	3	Negative

Regarding positive topics we have encountered that the most common comment was "Used the new options" and "Useful". Regarding both comments it can be seen on 6 of 7 evaluators, therefore we can confirm that is something that also results from the other explored topics had demonstrated (Questions and Heuristics), moreover, it confirms one of the initial hypothesis of this research. It is quite interesting that as it was denoted in previous sections, EV7 was one of the most critics evaluators, however, in the qualitative evaluation this evaluator has decided to choose both positive comments which contradicts what we found in quantitative data.

However, we encountered contradictions on the "Faster" and "Slower" topics where, for example, EV5 has chosen this options when it came to describe their experience, mentioning that the feeling was that it was faster in terms of answering the whole text but slower in terms of answering each question. Analyzing in detail their quotes and asking for clarification all the evaluates confirmed that the "Slower" topic was related to the increase on the number of answers which impacted at the beginning of the evaluation were they had to read several times the hint provided by the administrator of the test regarding when and where to use each one of the possible answers. According to their opinions it happened during almost half of the evaluation. Probably this problem it is also related to the burnout of the evaluation [36,39,45] and it can be one of the problems of this new proposals in detriment of their benefits.

6. Conclusions

This study presents an improved version of the original work done by Granollers [11] when it was created a new set of usability heuristics for general purpose and a new methodology to compute the Usability Percentage (UP) value, a quantitative value that tries to answer the typical question of “how usable is the system?”. Since then we have used this methodology regularly, so we have been getting to know its strengths and weaknesses, also identified through [34] to provide a novel method that improves the overall system and provides a better framework to be used for expert evaluators.

One of the main points to improve was the way to calculate the UP, since, although we know from the beginning that it is an indicative value, we want it to be as accurate as possible. To this end, we have proposed a new set of answers to the questions of the heuristics. The new set of answers has two main characteristics: on the one hand, the range of responses is greater, thus allowing better accuracy in the evaluators’ responses, and, on the other hand, answers are introduced to discern cases that the evaluator has not been able to perform.

The limitations of this study were related to the fact that the evaluation has been done to an already evaluated system. Though this was done on purpose so that the results between the two studies could be compared, it was noticed on the qualitative data that the evaluators still had in mind some problems that they encountered in the evaluation and were even more critics with the overall evaluation. Another key aspect is that we have applied this methodology to only one website, thus to validate it properly further research must be done. Also, the fact that it has been used the same experts as evaluators might be counterproductive and other set of expert must be selected.

Apart from this, we have taken advantage of the experimentation to try to improve the interpretation of the calculations, which is explained in Sections 4 and 5. To realize the experimentation we needed to be able to compare results, so we have based ourselves on all the same parameters of the previous study.

In terms of the results obtained, we consider important to highlight two aspects as the main scientific contributions: first, as explained in Section 5.3, the new set of answers provides greater consistency to the result of the UP, which gives it greater credibility. And, second, having the values of the UP with the top and lower percentages indicating the margin of error we have for each heuristic is of great value, it maximizes the degree of reliability of the value of the UP, that is very relevant when interpreting the results.

As future work, we think that for greater validation of results it will be necessary to perform new evaluations applied to other systems and consider to compare their results between them and also analyze from the same perspectives that we have been presented in this study. Future research will be attached to this point as we have to analyze the properties and results produced from other sources and understanding how the results differ between studies and also how this methodology will give a valuable results for the evaluators that uses this method. Moreover we have some insights that we would like to investigate like the results from Fig. 3 where it was explained that the initial heuristics evaluated have better results than the last ones. It will be necessary to also provide tests sorting randomly the heuristics to find if exists some kind of burning out as collected data from our evaluators suggests.

Another line of work, already started but not explained here because it does not yet have publishable results, is how to improve usability assessment through new Artificial Intelligence (AI) techniques. Our lines of work and expertise make us cautious with usability evaluations carried out entirely by AI programs. On the other hand, we see many possibilities in the use of AI systems trained with mathematical mechanisms and formulas such as those exposed in this work. So, one of the next steps will be to start working on this line.

Finally, regarding the platform that we have used to perform the tests it has been encountered some utilities that might help the development of the test like the automation of the figures and graphics that allows the analysis of the data as it has been presented in this research. Additionally, having a system that allows to export to LaTeX will help to boost the work done by researchers and will help to find the tool more useful for the community.

Abbreviations

The following abbreviations are used in this manuscript:

RUXAILAB	Remote User eXperience Artificial Intelligence LABoratory
UP	Usability Percentage
HE	Heuristic Evaluation
HUP	Heuristic Usability Percentage
STD	Standard Deviation
PD	Percentage Difference
ITC	Impossible to Check
IQR	Interquartile Range
EV	Evaluator

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CRediT authorship contribution statement

Marc Gonzalez Capdevila: Writing – original draft, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Karine Aparecida Pistili Rodrigues:** Writing – original draft, Investigation, Formal analysis, Data curation. **Tales Augusto Sartório Furlan:** Methodology, Investigation, Data curation. **Toni Granollers:** Writing – review & editing, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Marc Gonzalez Capdevila reports financial support was provided by FACENS Faculty of Engineering of Sorocaba. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be open and available on a public repository. From one side we provide the link to the github repository to get the source code of the project. Also we shared the evaluation made it public.

Acknowledgments

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Appendix A

See Table A.1.

Appendix B

See Table B.2.

Table A.1

Heuristic topics and questions.

Id	Heuristic	Questions
H1	Visibility and system state	- Does the application include a visible title page, section, or site? - Does the user always know where it is located? - Does the user always know what the system or application is doing? - Are the links clearly defined? - Can all actions be visualized directly? (No other actions are required)
H2	Connection between the system and the real world	- Does information appear in a logical order for the user? - Does the design of the icons correspond to everyday objects? - Does every icon do the action that you expect? - Does the system use phrases and concepts familiar to the user?
H3	User control and freedom	- Is there a link to come back to the initial state or homepage? - Are the functions “undo” and “re-do” implemented? - Is it easy to come back to an earlier state of the application?
H4	Consistency and standards	- Do link labels have the same names as their destinations? - Do the same actions always have the same results? - Do the icons have the same meaning everywhere? - Is the information displayed consistently on every page? - Are the colors of the links standard? If not, are they suitable for its use? - Do navigation elements follow the standards? (Buttons, check box,...)
H5	Recognition rather than memory, learning and anticipation	- Is it easy to use the system for the first time? - Is it easy to locate information that has already been searched for before? - Can you use the system at all times without remembering previous screens? - Is all content needed for navigation or task found in the “current screen”? - Is the information organized according to logic familiar to the end user?
H6	Flexibility and efficiency of use	- Are there keyboard shortcuts for common actions? - If there are, is it clear how to use them? - Is it possible to easily perform an action done earlier? - Does the design adapt to the changes of screen resolution? - Is the use of accelerators visible to the normal user? - Does it always keep the user busy? (without unnecessary delays)
H7	Help users recognize, diagnose and recover from errors	- Does it display a message before taking irreversible actions? - Are errors shown in real time? - Is the error message that appears easily interpretable? - Is some code also used to reference the error?
H8	Error prevention	- Does a confirmation message appear before taking the action? - Is it clear what information needs to be entered in each box on a form? - Does the search engine tolerate typos and spelling errors?
H9	Aesthetic and minimalist design	- Is used a design without redundancy of information? - Is the information short, concise, and accurate? - Is each item of information different from the rest and not confused? - Is the text well organized, with short sentences and quick to interpret?
H10	Help and documentation	- Is there the “help” option? - If so, is it visible and easy to access? - Is the help section aimed at solving problems? - Is there a section of frequently asked questions (FAQ)? - Is the help documentation clear, with examples?
H11	Save the state and protect the work	- Can users continue from a previous state (where they had previously been or from another device)? - Is “Autosave” implemented? - Does the system have a good response to external failures? (Power cut, internet not working,...)
H12	Color and readability	- Do the fonts have an adequate size? - Do the fonts use colors with sufficient contrast with the background? - Do background images or patterns allow the content to be read? - Does it consider people with reduced vision?
H13	Autonomy	- Does it keep the user informed of system status? - Moreover, is the system status visible and updated? - Can the user take their own decisions? (Personalization)
H14	Defaults	- Does the system or device give the option to return to factory settings? - If so, does it clearly indicate the consequences of the action? - Is the term “Default” used?
H15	Latency reduction	- Is the execution of heavy work transparent to the user? - While running heavy tasks, is remaining time or some animation shown?

Table B.2

Evaluators Comments.

Id	Heuristic	Questions
EV1	"I found the new methodology not very complicated to use since in the end it was based on an extension of the previous one so I have been faster than the first time. It is true that having more options when it comes to responding positively to the evaluation has helped me since it has allowed me to be more precise. Even so, I am not clear if I have correctly used the answers that were considered not to influence the calculation of the usability percentage. In any case I have used the new answer of impossible to check for some questions."	Faster/Not complicated/Used the new options/Useful/Precise
EV2	"In general I have not seen much difference between one method or another, although this may have been because I think I have not used the new options that have been presented to us since I have preferred to work in absolute thermoses for ease. Even so, comparing the two systems I have seen that the score has varied and I understand that this must be because I must have made modifications. Still, I think I answered it very quickly this time."	Not necessary/Not used the new options/Faster/Useful
EV3	"This is the second time I have made a review on this page and I think this has not helped me to be less critical. It has been difficult for me to understand the different options available and at the beginning I have had to read the meaning several times. It would still be a good option to use a different type of scale since I have felt like I waste time looking at the answers when I do not know exactly what answer to give to a specific topic. Yes, it is true that once I was halfway through, I hardly had to read the answers since I more or less remembered"	Not useful/Hard to understand/Slower/Used the new options
EV4	"I liked being able to have more options since it is something I was looking forward to from the first evaluation. The fact of having an answer that can be both positive and negative has surprised me and I think it helps to gauge the magnitude of the answers. Even so, for this analysis I have not found many questions that I have not been able to answer. I imagine that if a study has several problems, it must be quite useful."	Useful/Used the new options/Precise
EV5	"I liked to have more options of answers for each question, especially the new type of "Impossible to check", which helped on cases where I had doubts due to no being able to test a feature to its fullest. In my opinion the new answers are more clear to understand, making it easier to choose one of them, also it helped me to spend less time on the overall test. Although, I had to check the possible answers every time, as they are not very easy to memorize, which made me take more time to answer."	Not complicated/Used the new options/Useful/Faster/ Slower/Precise
EV6	"I do not think having that much answers helped at all. The only new answers that had impacted my evaluation was the new "Impossible to check", that I was able to use whenever I was no sure I could test something. It is was a little confusing and time demanding to choose from that many options, and I think it impacts on the overall result of evaluation, because evaluators will tend to answer more neutral answers instead of being more objective. Even though, I think it could be useful for some cases where the evaluation has to be very meticulous."	Hard to understand/Slower/Useful/Used the new options/Precise
EV7	"The new methodology is efficient for experienced evaluators and I think that re-evaluating the same site was useful, at least I went faster through the second heuristic after understanding the new answers. I believe that I was critic on both evaluations and I do not feel that I have used the new positive options. Having that many options allows being more specific, but that can be negative to less experienced evaluators, who would have more difficulty to choose between that many options. In the perspective of the calculation, I was not sure about the need of the "It is NOT a problem" type, it is considered a non countable question in the experiment, just like the "Not applicable", so evaluators already have an option for that case, which makes it very dispensable."	Hard to understand/Faster/Useful/Used the new options/Precise

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